LIFEGUARD: PERSONAL HEALTH & ENVIRONMENTAL SAFETY WEARABLE

Enhancing Personal Safety Through Integrated Sensor Fusion and Machine Learning

Authors: Evans Acheampong, Michael Kwabena Adu-Gyamfi

Supervisor: Dr. Percy Okae

Institution: Department of Computer Engineering, University of Ghana, Legon, Ghana

Emails: evansachie01@gmail.com, michaeladugyamfi76@gmail.com

Introduction

Wearable health monitoring has evolved significantly, yet most solutions focus solely on fitness tracking without integrating environmental safety. LifeGuard addresses critical gaps in personal safety for elderly individuals and industrial workers by combining real-time health monitoring with environmental hazard detection in a single, affordable device.

Using the Arduino Nicla Sense ME with 9 integrated sensors, our system provides fall detection (99.5% accuracy), activity recognition, and air quality monitoring—all at 60% lower cost than premium alternatives.

Problem Definition

- **Fragmented Systems:** Users require multiple devices for comprehensive monitoring, increasing cost and complexity
- **Limited Real-Time Response:** Most wearables lack immediate emergency alerts during falls or hazardous exposures
- **Accessibility Barriers:** Premium devices ($400-600) exclude vulnerable populations who need them most
- **Missing Integration:** Health and environmental data remain siloed, preventing holistic risk assessment"

Objectives

- Develop an integrated sensor fusion system combining physiological and environmental monitoring
- Implement real-time fall detection and activity recognition with >95% accuracy
- Create intelligent alert mechanisms for emergency response with <5 second latency
- Design accessible mobile and web interfaces for remote mon
- Achieve 72-hour battery life at 60% lower cost than competitors

System Design

WORKING SYSTEM OVERVIEW

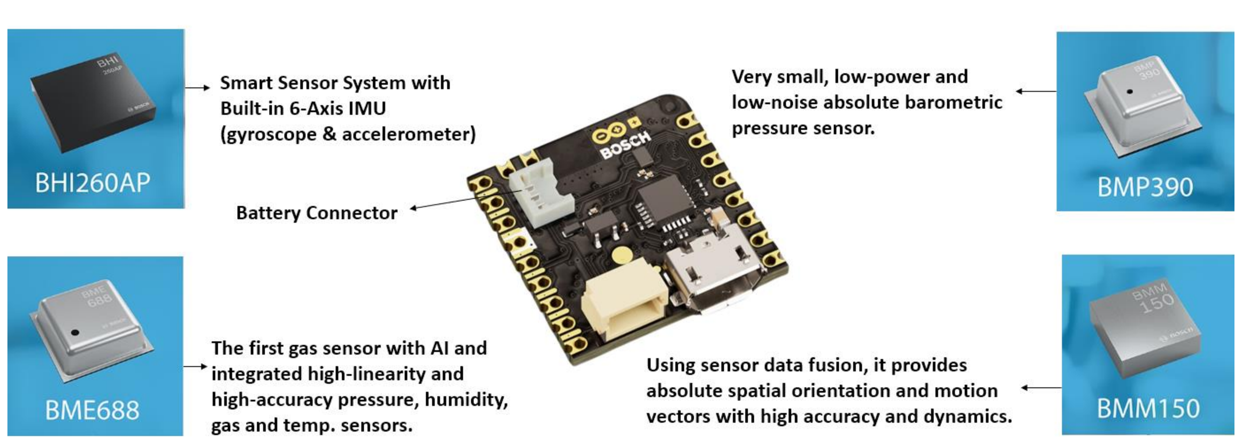

Figure 1: Working System Overview

SYSTEM ARCHITECTURE

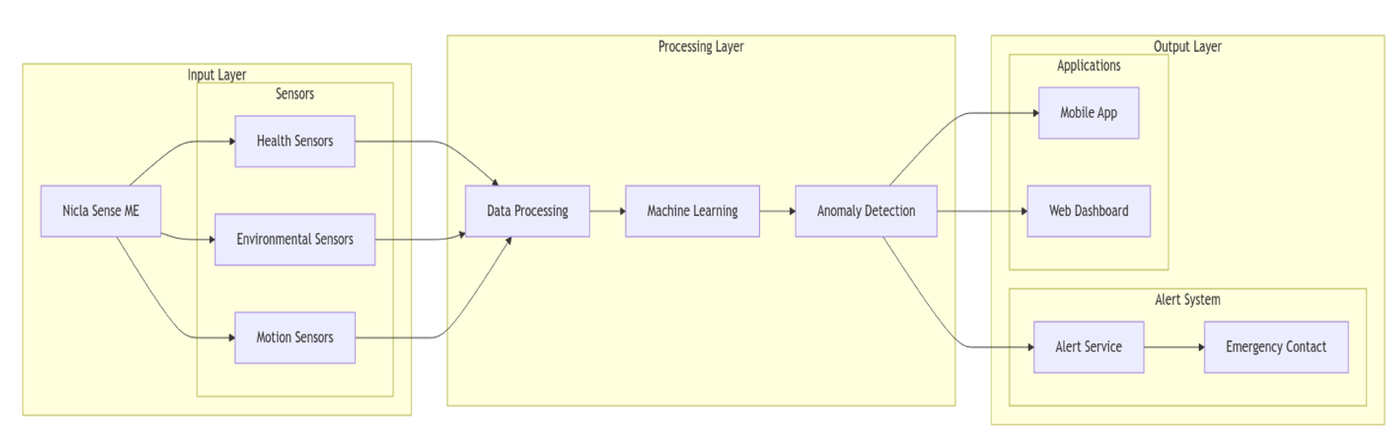

Figure 2: Proposed System Architecture

DATA FLOW DIAGRAM

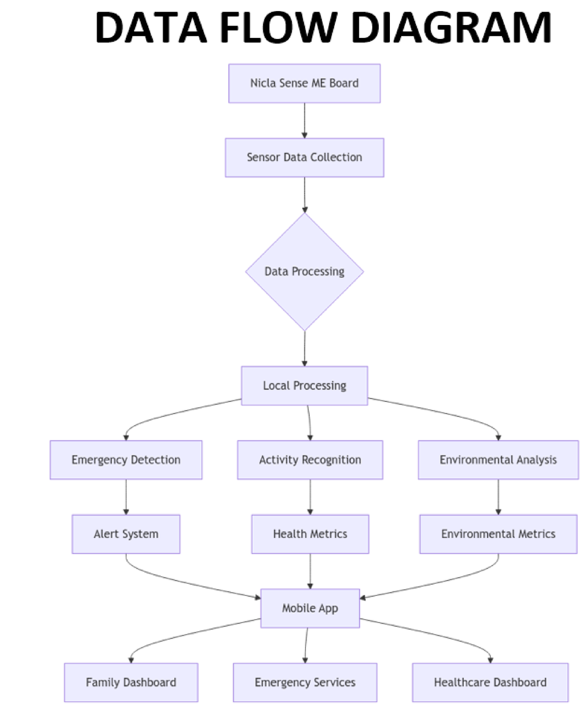

Figure 3: Proposed Data Flow Diagram

SYSTEM WIRING DIAGRAM

PICTORIAL VIEW OF SYSTEM OVERVIEW

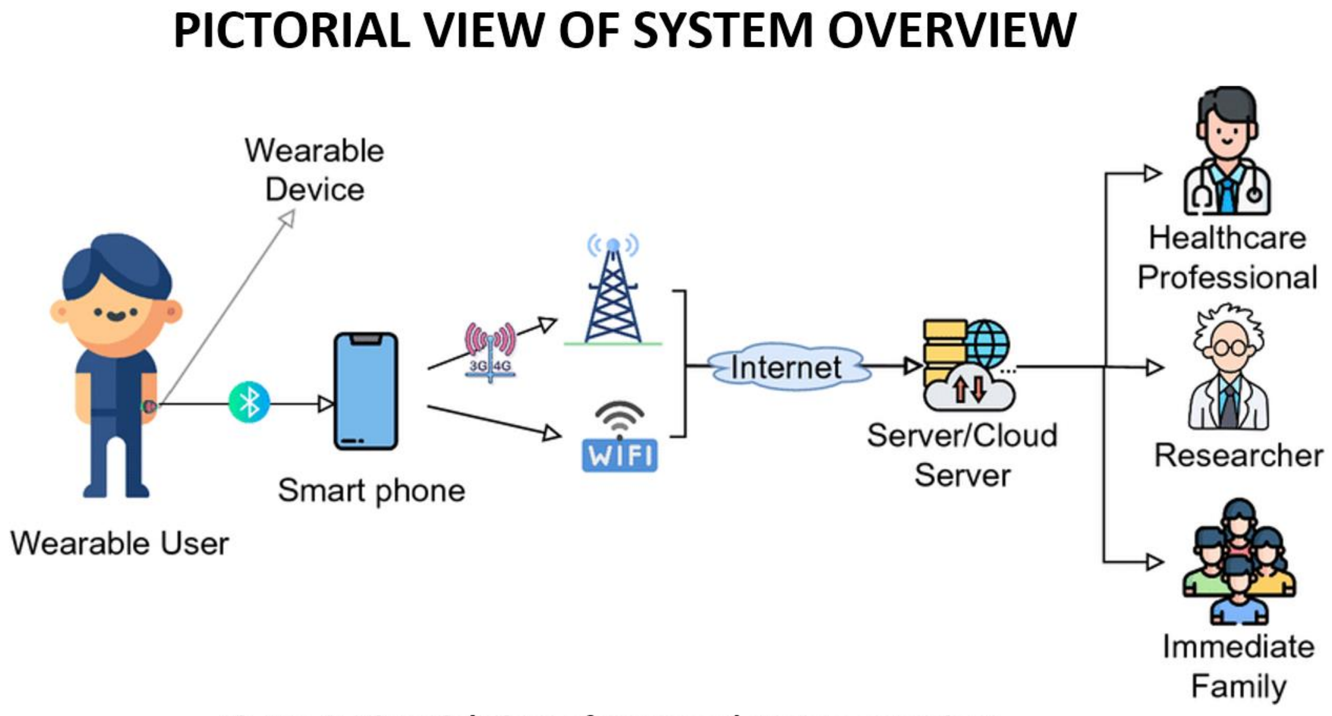

Figure 4: Pictorial view of proposed system overview

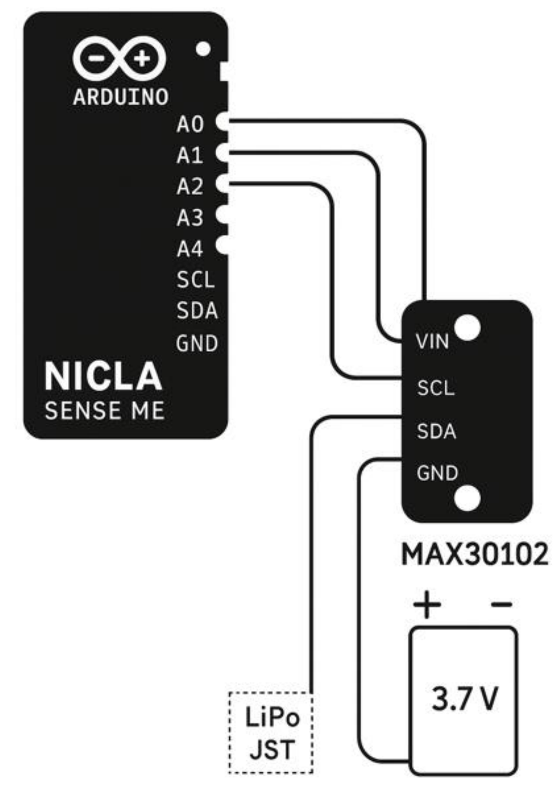

Figure 5: System Wiring Diagram

FINAL DEVICE DESIGN

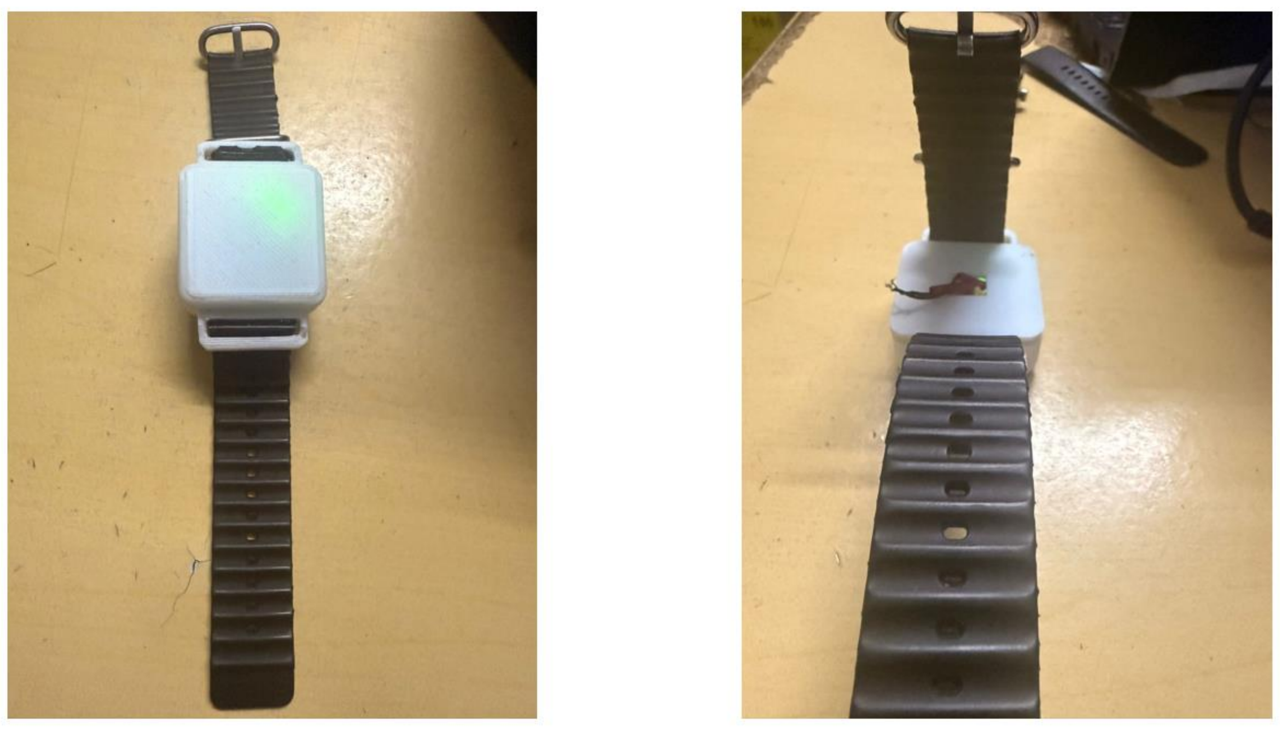

Figure 6: Final Device Design

System Development and Testing

The LifeGuard system was developed progressively through an iterative design process. Initial efforts focused on building the wearable enclosure, integrating sensors, and implementing embedded machine learning models for fall detection and activity recognition. Each module was tested independently to ensure accuracy and usability.

These components were later integrated with the mobile application and web dashboard, providing synchronized health and environmental monitoring. Final testing validated smooth interaction between the device, cloud database, and alert system, ensuring reliable performance in personal safety tracking and emergency response.

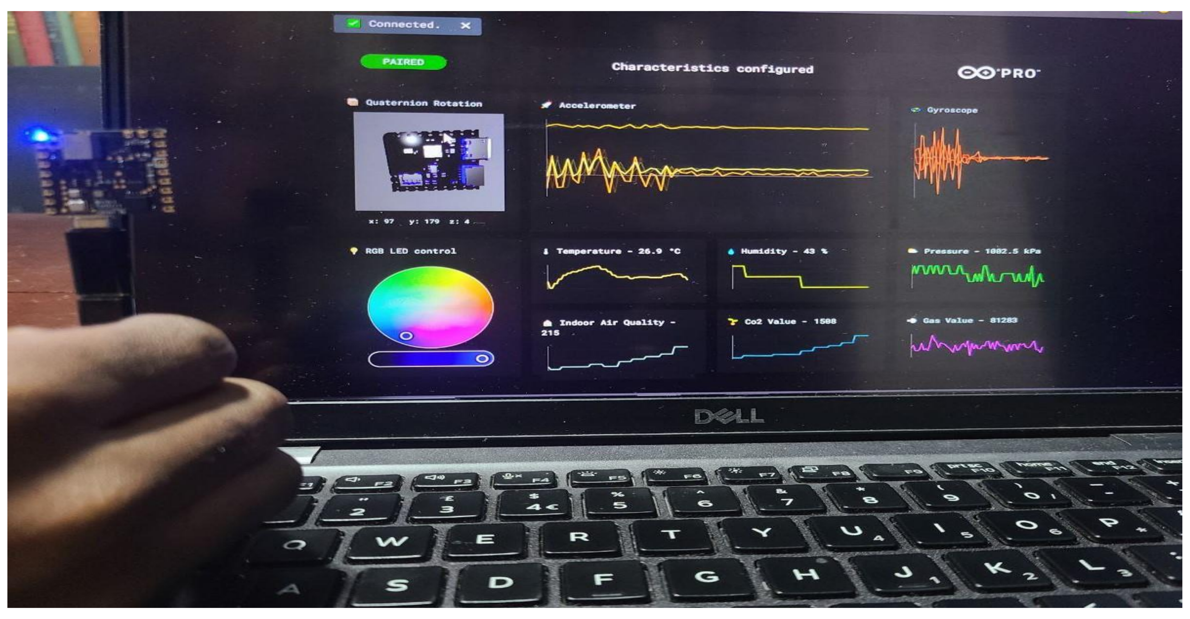

Figure 7: Live Sensor Data Testing

We contacted users using Google Forms once the system became live in order to get insightful input on their impressions of the tool. The purpose of the survey was to rate the system's usability, performance, usefulness of its features, and general satisfaction.

We sought to determine areas for development and make sure the system satisfied the demands of its users by gathering feedback the general public. Future platform updates and improvements were greatly influenced by the input provided.

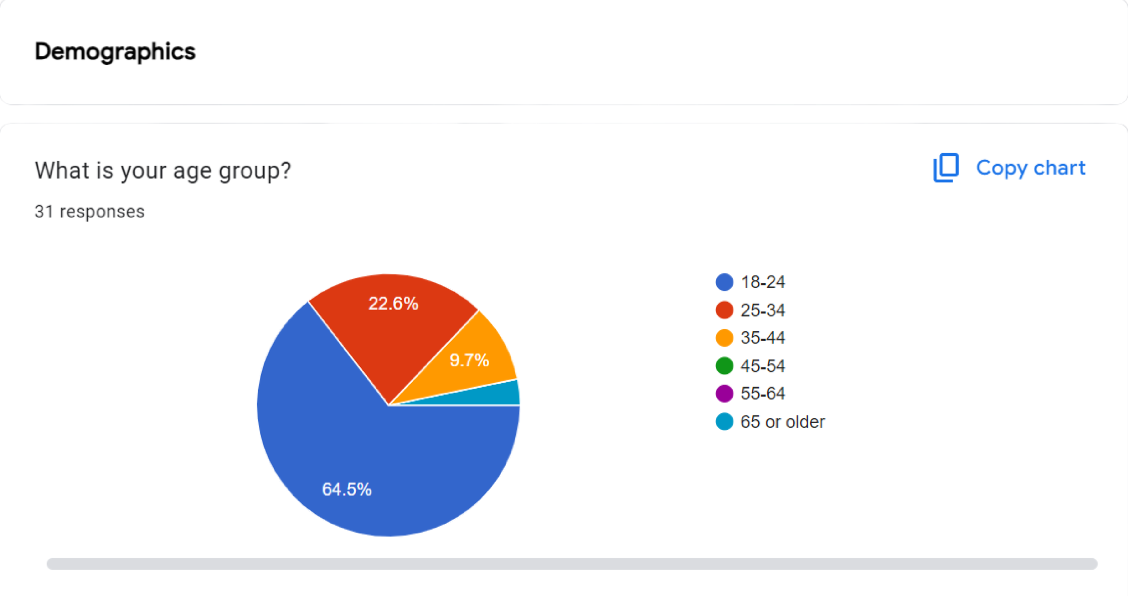

Figure 8: Demographics From Users

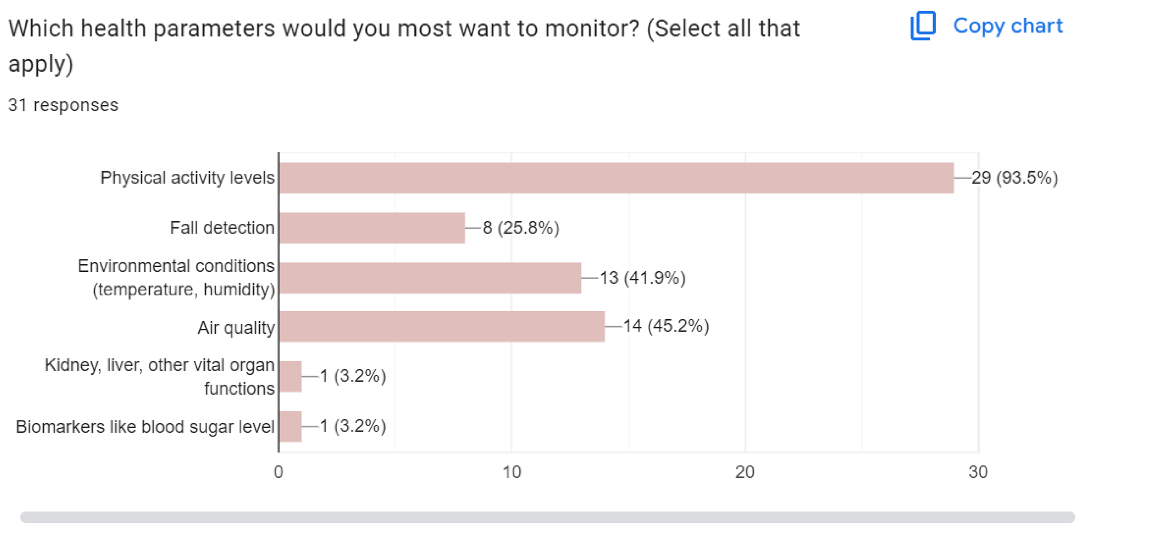

Figure 9: Results On what health parameter is needed most by users.

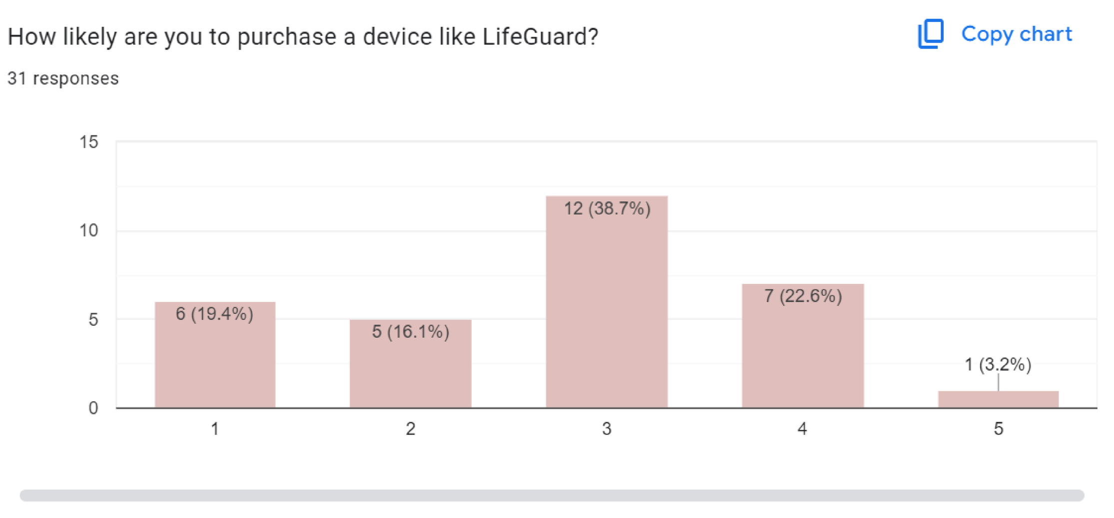

Figure 10:Results On How relevant the LifeGuard Platform was.

Conclusion

The project objectives for LifeGuard were successfully achieved. The wearable system integrates health and environmental sensing, real-time fall detection, and emergency alert functions into a compact, wrist-worn device. Survey results validated key design choices such as the wristwatch form factor, comprehensive monitoring features, and alert system reliability. The integration of a web dashboard and mobile app further enhanced accessibility and user engagement. Overall, LifeGuard meets the requirements of safety and preventive healthcare, providing an affordable and reliable solution for diverse user groups.

Future Extension

•Extending offline capabilities to ensure continuous monitoring without internet connectivity.
•Integrating advanced AI-driven health insights for more personalized safety and wellness recommendations.
•Optimizing battery health and rechargeability

References

[1] Hemapriya, D., Viswanath, P., & Mithra, V. M. (2019). Wearable medical devices: Design challenges and issues. *International Conference on Communication and Signal Processing (ICCSP)*, 0649-0653. IEEE. https://doi.org/10.1109/ICCSP.2019.8697969

[2] Zhu, T., Xu, S., Zhou, M., & Kan, H. (2024). Advances and perspectives in environmental health research in China. *Science of The Total Environment*, 906, 167558. https://doi.org/10.1016/j.scitotenv.2023.167558